



Datasets and Dataloaders

Rowel Atienza, PhD

University of the Philippines

github.com/roatienza

Updated: 2025

"The temptation to form premature theories upon insufficient data is the bane of our profession." - *Sherlock Holmes*

Outline

Dataset (Collection, Labelling, Data Structure)

Dataloader (APIs)

Datasets

$$\mathcal{D} = \{\mathbf{x}, \mathbf{y}\},$$
$$\mathbf{x} \in \mathbb{R}^{M \times N \times \dots}, \mathbf{y} \in \mathbb{R}^{S \times T \times \dots}$$

Stores the samples and their corresponding labels

Data Sources



Audio, Speech



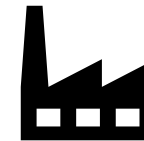
Image, Video



Sensors



Internet



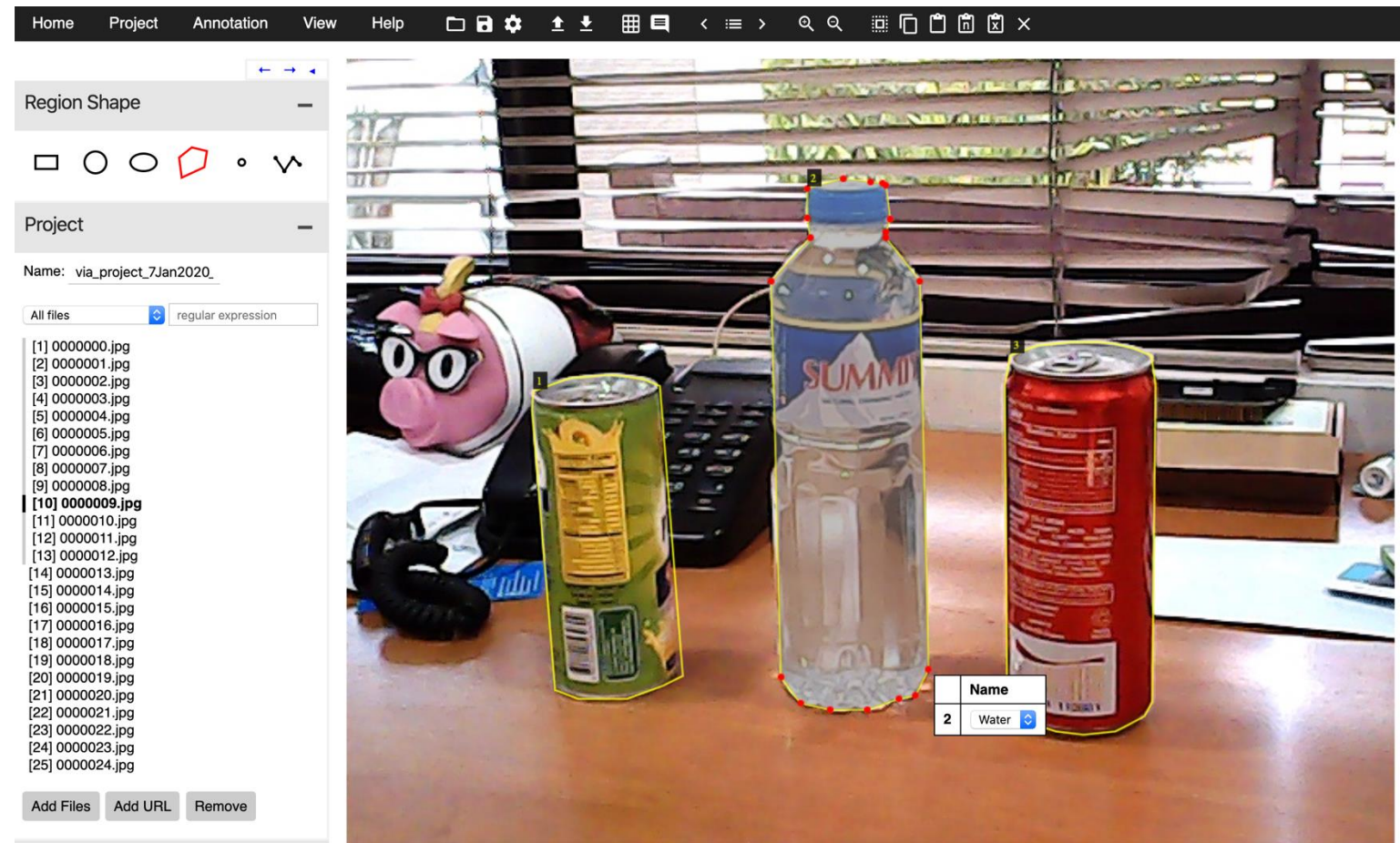
Business, Govt



Databases

Data Labelling or Annotation

VIA Annotation Tool



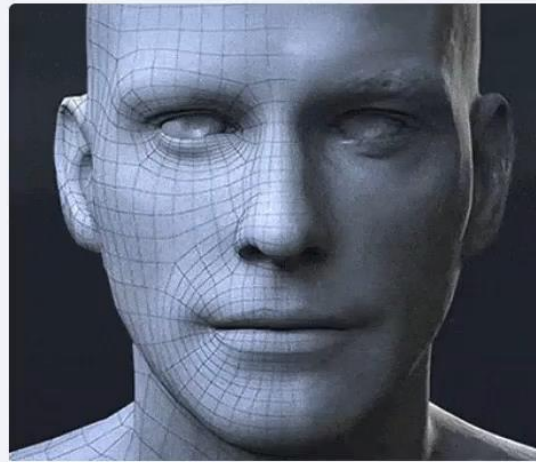
<https://www.robots.ox.ac.uk/~vgg/software/via/>



2D Bounding Boxes



Cuboid



Point & Landmark



Lines & Splines



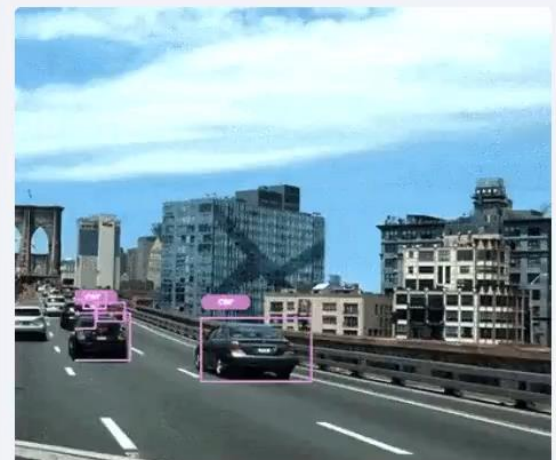
Text Annotation



Polygons



Semantic Segmentation



Video Annotation

GOO: A Dataset for Gaze Object Prediction in Retail Environments

Tomas, Reyes, Dionido, Ty,
Mirando, Casimiro, Atienza, Guinto

University of the Philippines & Samsung R&D PH

CVPR Workshops 2021

Publicly Available Datasets

HuggingFace Datasets <https://huggingface.co/datasets>



Torchvision, TorchAudio

CommonCrawl

Self-driving:

<http://apolloscape.auto/index.html>

APOLLOSCAPE

https://waymo.com/intl/en_us/dataset-download-terms/



Issues on datasets

Sufficiency

Train/Validation/Test

Bias

Sufficiency

The target test performance is achievable given a capable model

Train/Validation/Test Split

70/10/20

70/20/10

80/0/20

80/10/10

No hard rule but no duplication of samples in the splits!

Bias

Sampling, Exclusion, Racial, Measurement, Recall, Association, Observer

GOO: A Dataset for Gaze Object Prediction in Retail Environments

Tomas, Reyes, Dionido, Ty,
Mirando, Casimiro, Atienza, Guinto

University of the Philippines & Samsung R&D PH

CVPR Workshops 2021

<https://www.telusinternational.com/articles/7-types-of-data-bias-in-machine-learning>

PyTorch Datasets

<https://pytorch.org/vision/stable/datasets.html>

`torch.utils.data.Dataset`

Supports `__getitem__()` and `__len__()`

<https://pytorch.org/docs/stable/data.html#torch.utils.data.Dataset>

torchvision.datasets

<https://pytorch.org/vision/stable/datasets.html>

Common datasets for vision classification, detection, segmentation, action recognition, captioning

COCO 2017 Keypoint Detection Task



torchaudio.datasets

<https://pytorch.org/audio/stable/datasets.html>

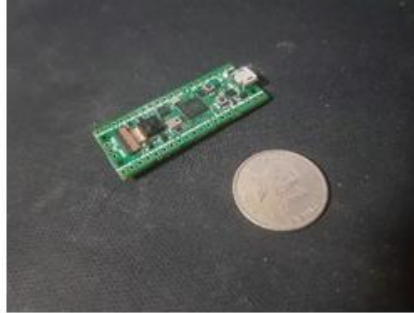
Common datasets for automatic speech recognition (ASR), text to speech (TTS), keyword spotting (KWS), voice cloning, music genre recognition

Objective:

- Detect spoken speech commands on a **Cortex-M0** microcontroller
- Use depth pruning to create optimized neural network for detection

Commands:

- Up, Down
- Left, Right
- On, Off
- Yes, No
- Go, Stop



Results:

Metric	Result	Improvement*
Model Size	43KB	1.24×
Inference Time	275ms	1.24×
Accuracy	90.77%	↓2.2%

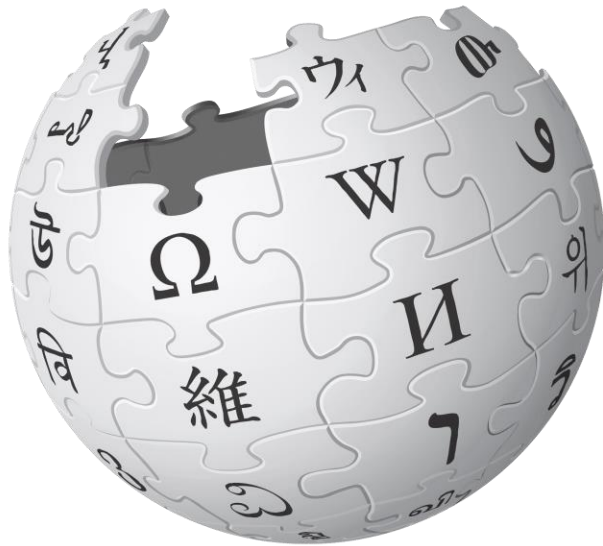
**compared to baseline KWS DS-CNN model*



torchtext.datasets

<https://pytorch.org/text/stable/datasets.html>

Common datasets for text classification, language modelling, machine translation, sequence tagging, question answer, unsupervised learning



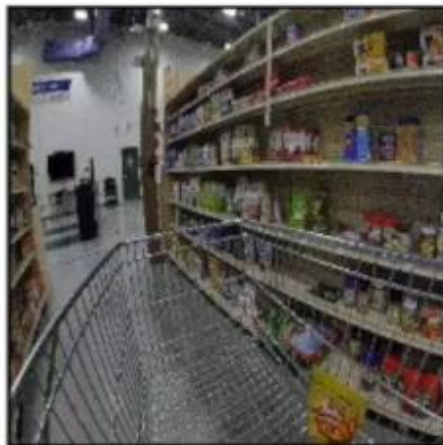
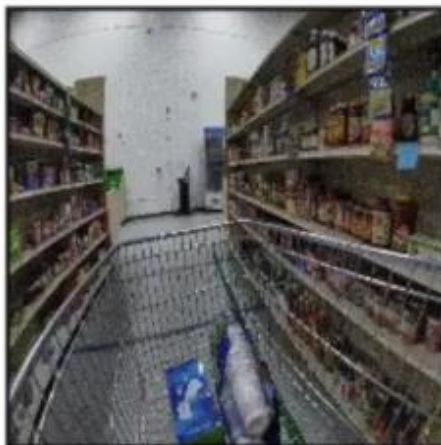
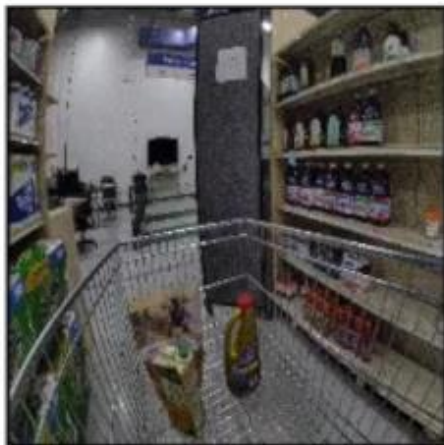
WIKIPEDIA
The Free Encyclopedia

Retail Vision Datasets

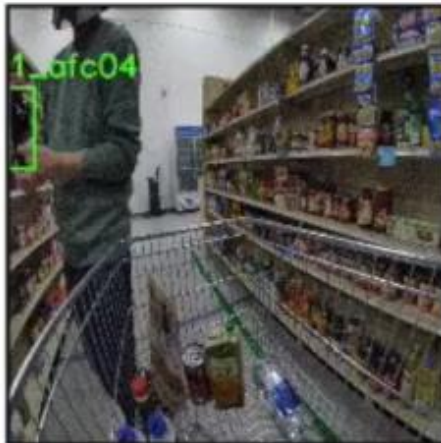
Grocery Vision Challenge

<https://grocery-vision.github.io/>

Takes



Returns



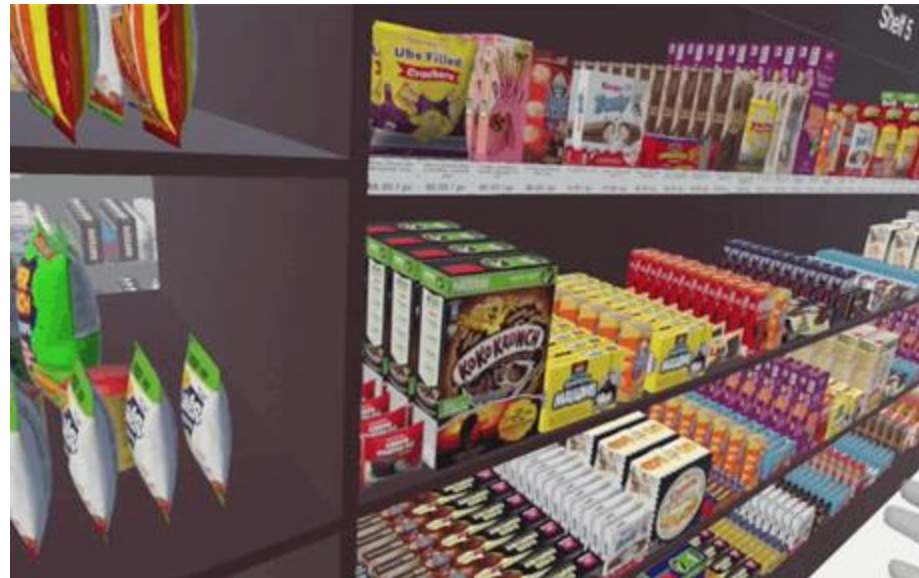
Rummage



Sari Sandbox

<https://sarisandbox.github.io/>





Talking Faces

- Crema and HDTF: Talking Faces Datasets

Visual (Stable Perspective)



Audio-Visual Matching



GT(left) vs Inferred (right)



Celebrity Datasets (Celeb A)

Sample Images

- 200k images
- 10,177 number of identities,
- 202,599 number of face images, and
- 5 landmark locations, 40 binary attributes annotations per image.

Eyeglasses



Bangs



Pointy Nose



Oval Face



Wearing Hat



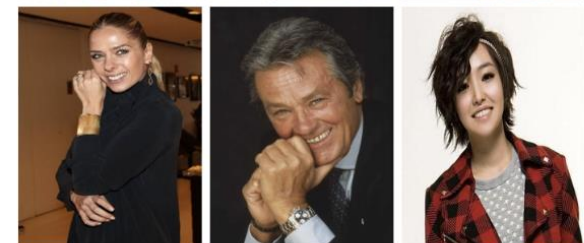
Wavy Hair



Mustache



Smiling



<https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>

Common Crawl



The Data ▼ Resources ▼

Overview

Over **300 billion** pages spanning 18 years.

Free and open corpus since 2007.

Cited in over **10,000** research papers.

3–5 billion new pages added each month.



WIKIPEDIA
The Free Encyclopedia

Generative AI Datasets

Wan

<https://github.com/Wan-Video/Wan2.1>

Input: First person view of a modern breakfast nook with food everywhere

Generative AI



Output

Dataloaders

https://pytorch.org/tutorials/beginner/basics/data_tutorial.html

Why dataloaders

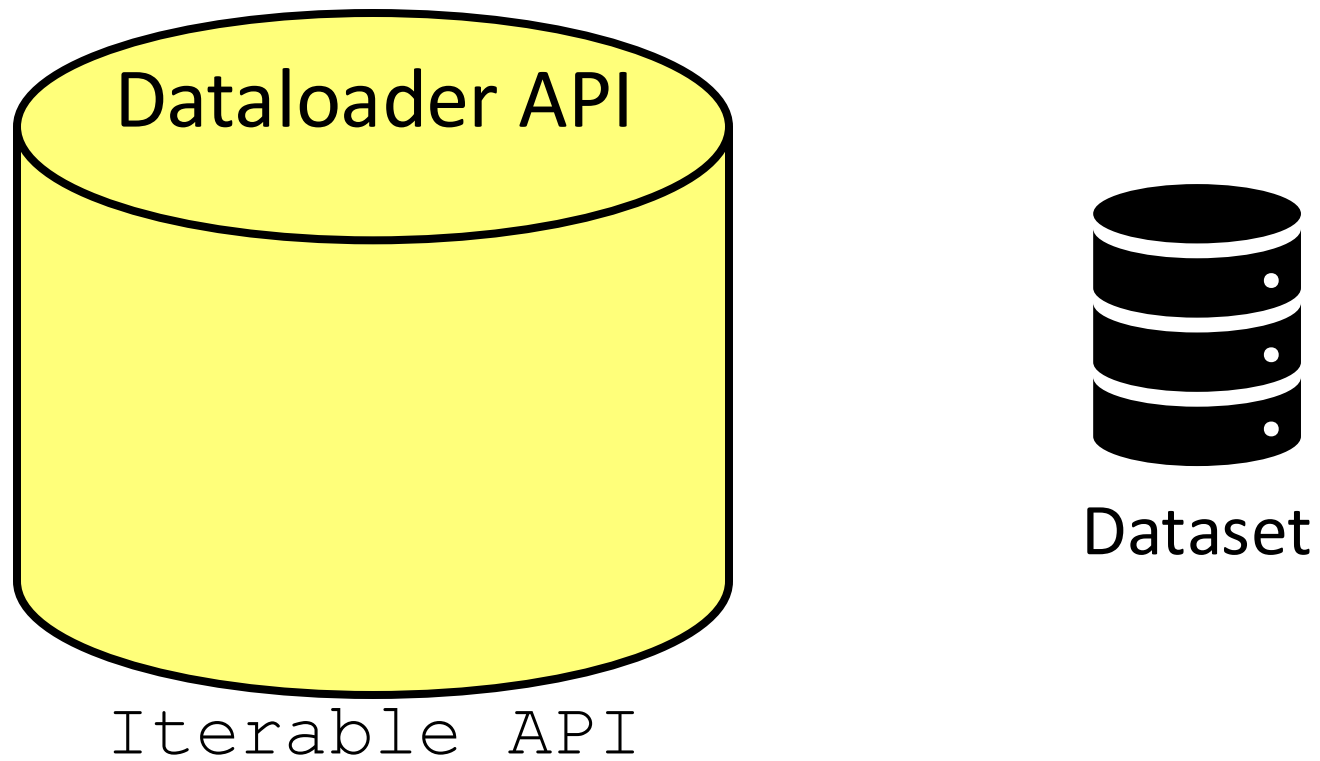
Mini-batch loading

Dataset shuffling

Multi-processing/threading

Data collating (pre-processing)

Dataloader is Dataset Wrapper



torch.utils.data.DataLoader

```
DataLoader(dataset,  
            batch_size=1,  
            shuffle=False,  
            num_workers=0,  
            collate_fn=None,  
            pin_memory=False,  
            ...)
```

```

def collate_fn(self, batch):
    mels = []
    labels = []
    wavs = []
    for sample in batch:
        waveform, sample_rate, label, speaker_id, utterance_number = sample
        # ensure that all waveforms are 1sec in length; if not pad with zeros
        if waveform.shape[-1] < sample_rate:
            waveform = torch.cat([waveform, torch.zeros((1, sample_rate - waveform.shape[-1]))], dim=-1)
        elif waveform.shape[-1] > sample_rate:
            waveform = waveform[:, :sample_rate]

        # mel from power to db
        mels.append(torch.tensor(librosa.power_to_db(self.transform(waveform).squeeze().numpy(), ref=np.max)))
        labels.append(torch.tensor(self.class_dict[label]))
        wavs.append(waveform)

    mels = torch.stack(mels)
    labels = torch.stack(labels)
    wavs = torch.stack(wavs)

    return mels, labels, wavs

```

End